

Inverse function – year 11

Given that $f(x) = e^x$ then $f'(x) = e^x$.
(e is a constant $e = 2.71828 \dots$)

1. Find the largest possible domain for which the following are monotonic increasing functions.
 - a. $f(x) = \sqrt{4 - x^2}$
 - b. $f(x) = \sqrt{x^2 - 4}$
 - c. $f(x) = -\frac{1}{x+2}$
 - d. $f(x) = 3x - x^2$
 - e. $f(x) = x^2 + 6x + 8$
2. Sketch the graph of each function. Then use reflection in the line $y = x$ to sketch the inverse relation. State whether or not the inverse is a function, and find its equation if it is. Also, state whether $f(x)$ and $f^{-1}(x)$ (if it exists) are monotonic increasing, decreasing or neither.
 - a. $f(x) = 2x$
 - b. $f(x) = x^3 + 1$
 - c. $f(x) = \sqrt{1 - x^2}$
 - d. $f(x) = x^2 - 4$
 - e. $f(x) = 2^x$
 - f. $f(x) = \sqrt{x - 3}$
3. A function is defined as $f(x) = 1 + e^{2x}$. Find the inverse function $f^{-1}(x)$ and state the domain and range.
4. Given that $f(x) = \log_e \left(\frac{1+x}{1-x} \right)$. Find the equation for the inverse function $f^{-1}(x)$.
5. Restrict the domain of the following function so that it has an inverse that is a function. Then sketch the restricted function and its inverse:
 $y = \sin \left(\frac{\pi x}{2} \right)$.
6. Find the inverse function and state the domain and range of the inverse: $f(x) = \log_e(x + 1)$
7. Given that $y = x^3 - 9x + 6\sqrt{3}$
 - a. Show that it has turning points at $(\sqrt{3}, 0)$ and $(-\sqrt{3}, 12\sqrt{3})$.
 - b. Find the other point where the graph cuts the x -axis.
 - c. Let $f(x) = x^3 - 9x + 6\sqrt{3}$ for the restricted domain $-\sqrt{3} \leq x \leq \sqrt{3}$. Sketch the graph of $y = f(x)$ and explain why the inverse function $f^{-1}(x)$ exists.
- d. State the domain and range of $f^{-1}(x)$.
- e. Sketch $y = f^{-1}(x)$.
- f. Find the equation of the tangent to the curve $y = f^{-1}(x)$ at the point where the curve cuts the x -axis.
8. Given $y = e^x - 1$
 - a. Show that it is a monotonic increasing function.
 - b. Show that the graph of $y = e^x - 1$ has a gradient of 1 at the point $(0, 0)$.
 - c. Carefully sketch $y = e^x - 1$, $y = x$ and the inverse function on the same diagram.
 - d. Show that the inverse function is $y = \log_e(x + 1)$
 - e. Solve $\log_e(x + 1) < x$.
9. The function $f(x) = x + 3$ is defined over the domain $0 \leq x \leq 2$.
 - a. State the range of $f(x)$.
 - b. State the domain and range of $f^{-1}(x)$.
 - c. Write down the rule for $f^{-1}(x)$.
10. The function F is defined by $F(x) = \sqrt{x}$ over the domain $0 \leq x \leq 4$.
 - a. State the range of $F(x)$.
 - b. State the domain and range of $F^{-1}(x)$.
 - c. Write down the rule for $F^{-1}(x)$.
 - d. Graph F and F^{-1}
11. Consider the functions $f(x) = 3x + 2$ and $g(x) = \frac{1}{3}(x - 2)$
 - a. Find $f(g(x))$ and $g(f(x))$.
 - b. What is the relationship between $f(x)$ and $g(x)$?
12. Each function $g(x)$ is defined over a restricted domain so that $g^{-1}(x)$ is a function. Find $g^{-1}(x)$ and write down its domain and range.
 - a. $g(x) = x^2, x \geq 0$
 - b. $g(x) = x^2 + 2, x \leq 0$
 - c. $g(x) = -\sqrt{4 - x^2}, 0 \leq x \leq 2$

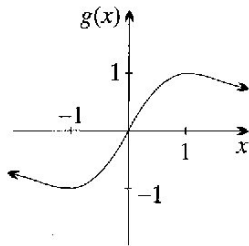
13. Given the function $y = x^3 - 1$
- Write down $\frac{dy}{dx}$
 - Make x the subject and hence find $\frac{dx}{dy}$
 - Hence show that $\frac{dy}{dx} \times \frac{dx}{dy} = 1$
14. Repeat the previous question for $y = \sqrt{x}$
15. The function $F(x) = x^2 + 2x + 4$ is defined over the domain $x \geq -1$
- Sketch the graphs of $F(x)$ and $F^{-1}(x)$ on the same diagram.
 - Find the equation of $F^{-1}(x)$ and state its domain and range.
16. Given that $f(x) = 1 - \log_e x$.
- Solve $f(x) = 0$.
 - Sketch the graph of $f(x) = 1 - \log_e x$. Hence sketch the graph of $f^{-1}(x)$ on the same diagram.
 - Find the equation of $f^{-1}(x)$ and state its domain and range.
 - Classify $f(x)$ and $f^{-1}(x)$ as increasing, decreasing or both.
17. Given $g(x) = \frac{x+2}{x+1}$, for $x > -1$.
- Sketch the function
 - Find $g^{-1}(x)$ and sketch it on the same diagram. Is $g^{-1}(x)$ increasing or decreasing?
 - Find any values of x for which $g(x) = g^{-1}(x)$.
18. The previous question seems to imply that the graphs of a function and its inverse can only intersect on the line $y = x$. This is not always the case.
- Find the equation of the inverse of $y = -x^3$
 - At what points do the graphs of the functions and its inverse meet?
 - Sketch the situation.
19. Given $f(x) = x^2$ and $g(x) = (x+2)^2 - 4$.
- Explain how the graph of $f(x) = x^2$ must be transformed to obtain the graph of $g(x) = (x+2)^2 - 4$.
- Hence sketch the graph of $g(x)$, showing the x and y intercepts and the vertex.
 - What is the largest domain containing $x = 0$ for which $g(x)$ has an inverse function?
 - Let $g^{-1}(x)$ be the inverse function corresponding to the domain of $g(x)$ in part c. What is the domain of $g^{-1}(x)$? Is $g^{-1}(x)$ increasing or decreasing?
 - Find the equation of $g^{-1}(x)$, and sketch it on your diagram in part b.
 - Classify $g(x)$ and $g^{-1}(x)$ as either monotonic increasing, decreasing or neither.
20. $F(x) = x^3 - 3x$
- Show that it is an odd function.
 - Sketch the graph of $F(x)$, showing the x -intercepts and the coordinates of the two stationary points. Is $F(x)$ monotonic increasing or decreasing?
 - What is the largest domain containing $x = 0$ for which $F(x)$ has an inverse function?
 - State the domain of $F^{-1}(x)$, and sketch it on the same diagram as part b.
21. Given that $f(x) = \frac{e^x}{1+e^x}$
- State the domain.
 - Show that $f'(x) = \frac{e^x}{(1+e^x)^2}$.
 - Hence explain why $f(x)$ is increasing for all x .
 - Explain why $f(x)$ has an inverse function, and find its equation.
22. Given $f(x) = \frac{1}{1+x^2}$.
- Sketch $y = 1 + x^2$ and hence sketch $f(x) = \frac{1}{1+x^2}$. Is $f(x)$ monotonic increasing or decreasing?
 - What is the largest domain containing $x = -1$ for which $f(x)$ has an inverse function?
 - State the domain of $f^{-1}(x)$, and sketch it on the same diagram as part a.
 - Find the rule for $f^{-1}(x)$.
 - Is $f^{-1}(x)$ increasing or decreasing?
23. Linear function $f(x) = mx + b$
- Show that any linear function $f(x) = mx + b$ has an inverse function if $m \neq 0$

- b. Does the constant function $F(x) = b$ have an inverse function?

24. Given $f(x) = x - \frac{1}{x}$, for $x > 0$.

- Sketch $y = f(x)$, for $x > 0$
- Sketch $y = f^{-1}(x)$ on the same diagram.
- By completing the square or using the quadratic formula, show that $f^{-1}(x) = \frac{1}{2}(x + \sqrt{4 + x^2})$.

25. The diagram shows the function $g(x) = \frac{2x}{1+x^2}$, whose domain is all real x .



- Show that $g\left(\frac{1}{a}\right) = g(a)$, for all $a \neq 0$
- Hence explain why the inverse of $g(x)$ is not a function
- i. What is the largest domain of $g(x)$ is not a function for which $g^{-1}(x)$ exists?

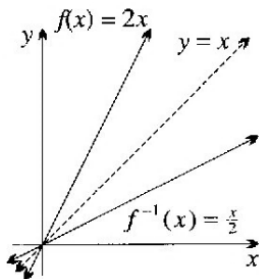
- Sketch $g^{-1}(x)$ for this domain of $g(x)$.
 - Find the equation of $g^{-1}(x)$ for this domain of $g(x)$
- d. Repeat part c for the largest domain of $g(x)$ that does not contain $x = 0$.
- e. Show that the two expressions for $g^{-1}(x)$ in part c and d are reciprocals of each other. Why could we have anticipated this?

26. Consider the function $f(x) = \frac{1}{6}(x^2 - 4x + 24)$.

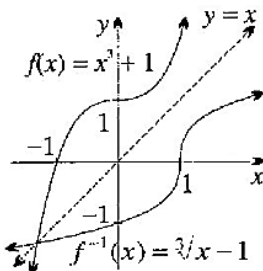
- Sketch the parabola $y = f(x)$, showing the vertex and any x - or y -intercepts
- State the largest domain contain only positive number for which $f(x)$ has an inverse function $f^{-1}(x)$.
- Sketch $f^{-1}(x)$ on your diagram from part a, and state its domain.
- Find any points of intersection of the graphs of $y = f(x)$ and $y = f^{-1}(x)$.
- Let N be a negative real number. Find $f^{-1}(f(N))$.

Answer

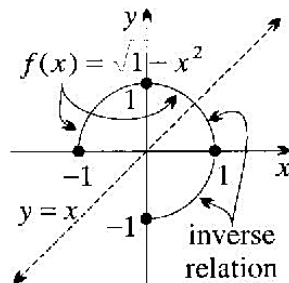
- a. $-2 \leq x \leq 0$; b. $x \geq 2$; c. $x < -2$, $x > -2$; d. $x \leq 1.5$; e. $x \geq -3$.
- a. $f^{-1}(x) = \frac{1}{2}x$, both increasing;



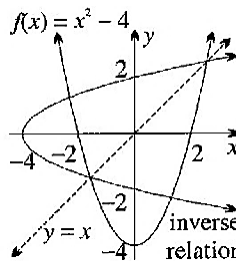
- b. $f^{-1}(x) = \sqrt[3]{x-1}$, both increasing;



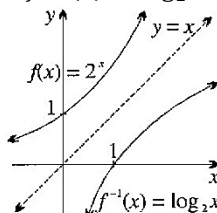
- c. The inverse is not a function, f is neither increasing nor decreasing;



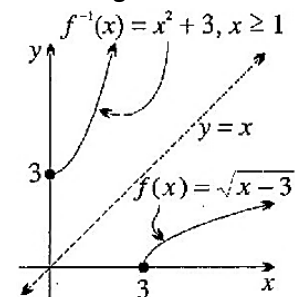
- d. The inverse is not a function. f is neither increasing nor decreasing;



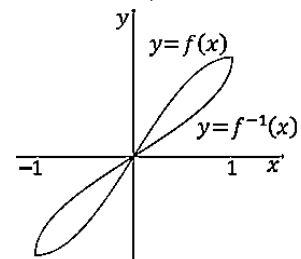
- e. $f^{-1}(x) = \log_2 x$, both increasing;



- f. $f^{-1}(x) = x^2 + 3$, $x \geq 0$, both increasing.

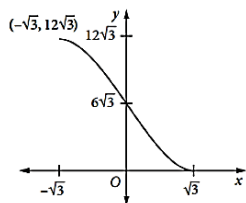


- $f^{-1}(x) = \frac{1}{2} \log_e(x-1)$, $D: x > 1$, $R: y \in \mathbb{R}$
- $f^{-1}(x) = \frac{e^x - 1}{e^x + 1}$
- $-1 \leq x \leq 1$,



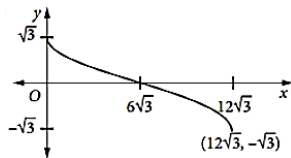
- $f^{-1}(x) = e^x - 1$, all real x , $y > -1$
- b. $(-2\sqrt{3}, 0)$;

c.



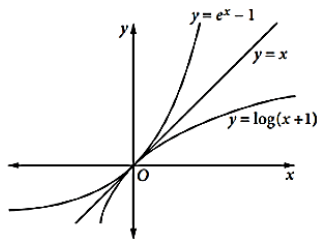
d. $0 \leq x \leq 12\sqrt{3}$, $-\sqrt{3} \leq y \leq \sqrt{3}$;

e.



f. $x + 9y = 6\sqrt{3}$

8. c.

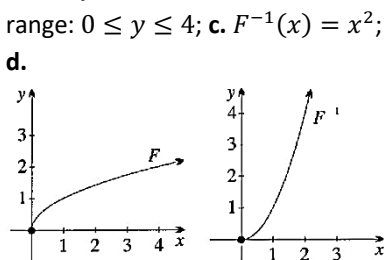


e. $-1 < x < 0$, $x > 0$

9. a. $3 \leq y \leq 5$; b. domain: $3 \leq x \leq 5$, range: $0 \leq y \leq 2$; c. $f^{-1}(x) = x - 3$

10. a. $0 \leq y \leq 2$; b. domain: $0 \leq x \leq 2$, range: $0 \leq y \leq 4$; c. $F^{-1}(x) = x^2$;

d.



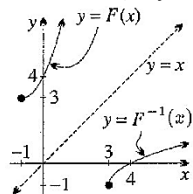
11. a. Both x ; b. They are inverse functions.

12. a. $g^{-1}(x) = \sqrt{x}$, domain: $x \geq 0$, range: $y \geq 0$; b. $g^{-1}(x) = -\sqrt{x-2}$, domain: $x \geq 2$, range: $y \leq 0$; c. $g^{-1}(x) = \sqrt{4-x^2}$, $-2 \leq x \leq 0$, domain: $-2 \leq x \leq 0$, range: $0 \leq y \leq 2$

13. a. $3x^2$; b. $\frac{1}{3}(y+1)^{-\frac{2}{3}}$

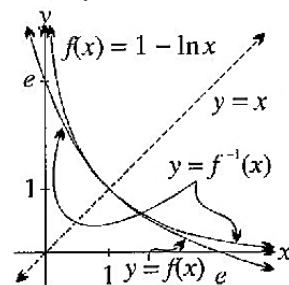
14. $\frac{dy}{dx} = \frac{1}{2\sqrt{x}}$, $\frac{dx}{dy} = 2y$

15. b. $F^{-1}(x) = -1 + \sqrt{x-3}$, domain: $x \geq 3$, range: $y \geq -1$



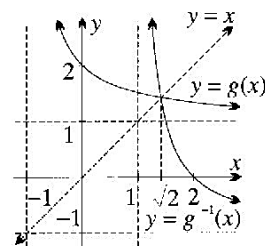
16. a. $x = e$; b. Reflect $y = \ln x$ in the x -axis, then shift it one unit up; d.

$f^{-1}(x) = e^{1-x}$, domain: all real x , range: $y > 0$; e. Both are decreasing.



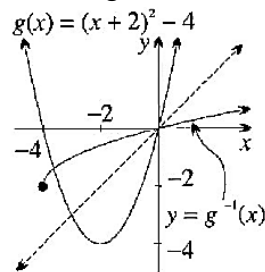
17. b. $g^{-1}(x) = \frac{2-x}{x-1}$, for $x > 1$,

decreasing; c. $x = \sqrt{2}$. It works because the graphs meet on the line of symmetry $y = x$.

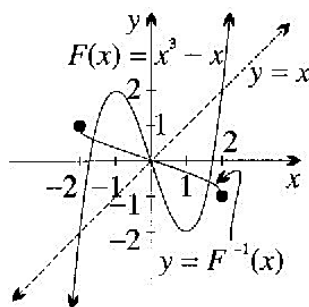


18. a. $y = \sqrt[3]{-x}$; b. $(-1, 1)$, $(0, 0)$ and $(1, -1)$

19. a. Shift two units to the left and four units down; b. x -intercepts: $-4, 0$, vertex: $(-2, -4)$; c. $x \geq -2$; d. $x \geq -4$, increasing; e. $g^{-1}(x) = -2 + \sqrt{x+4}$; f. $g(x)$ is neither, $g^{-1}(x)$ is increasing.



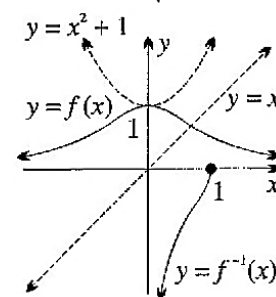
20. b. x -intercepts: $0, \sqrt{3}, -\sqrt{3}$, stationary points: $(-1, 2)$, $(1, -2)$, neither increasing nor decreasing; c. $-1 \leq x \leq 1$; d. $-2 \leq x \leq 2$



21. a. all real x ; c. $f'(x) > 0$ for all real x ; d. For each value of y , there is only one value of x . That is, the graph of $f(x)$ passes the horizontal line test. $f^{-1}(x) = \log\left(\frac{x}{1-x}\right)$

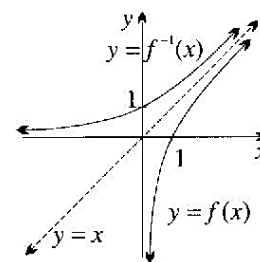
22. a. neither; b. $x \leq 0$; c. $0 < x \leq 1$; d.

$f^{-1}(x) = -\sqrt{\frac{1-x}{x}}$; e. increasing

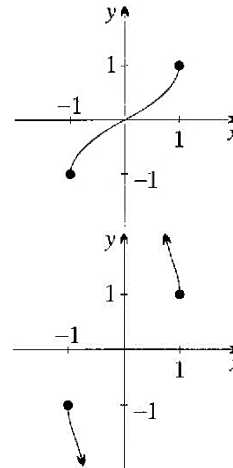


23. b. No. The graph of inverse is a vertical line, which is not a function.

24. a. b.



25. b. From part a. we see, for example, that $g\left(\frac{1}{2}\right) = g(2)$, so the inverse is not a function; c. i. $-1 \leq x \leq 1$, iii. $g^{-1}(x) = \frac{1-\sqrt{1-x^2}}{x}$; d. domain: $x \leq -1$ or $x \geq 1$, $g^{-1}(x) = \frac{1+\sqrt{1-x^2}}{x}$; e. Because of the result in part a.



26. a. vertex: $\left(2, \frac{10}{3}\right)$, y -intercept: 4; b. $x \geq 2$; c. $x \geq \frac{10}{3}$; d. The easy way is to solve $y = f(x)$ simultaneously with $y = x$. The intersect at $(4, 4)$ and $(6, 6)$; e. $4 - N$

